

# TRASH CAT: ENDLESS RUNNER

## COLLEGE PROJECT DOCUMENTATION (FORMAL VERSION)

---

### 1. ABSTRACT

This document presents the design and development of **Trash Cat: Endless Runner**, a 2D/2.5D mobile and desktop game developed using Unreal Engine 5. The project focuses on implementing core game development concepts such as procedural level generation, player control systems, collision handling, scoring mechanisms, and cross-platform deployment.

The primary objective of the game is to control a cat character that continuously runs through an environment while collecting items and avoiding obstacles. The project demonstrates practical knowledge in game engine usage, asset pipeline integration, optimization, and Android build configuration.

---

### 2. INTRODUCTION

Game development combines programming, design, and art to create interactive experiences. Endless runner games are widely used as beginner-to-intermediate projects due to their simple concept and scalable complexity.

In this project, an endless runner game titled **Trash Cat** was developed. The game involves a character that automatically runs forward, and the player must control its movement to avoid obstacles and collect rewards.

This project was developed as part of academic learning and serves as a portfolio-ready implementation of real-world game development workflows.

---

### 3. OBJECTIVES

The main objectives of this project are:

- To understand Unreal Engine workflow
  - To implement core gameplay mechanics
  - To design a functional endless runner system
  - To handle asset import and conversion
  - To deploy the game on Android and Windows platforms
  - To implement data persistence (high score system)
-

## 4. GAME OVERVIEW

### 4.1 Title

**Trash Cat: Endless Runner**

### 4.2 Genre

Endless Runner

### 4.3 Target Platforms

- Windows 7 / 8 / 10 / 11
- Android 8.0 and above

### 4.4 Game Description

The player controls a cat that runs endlessly in a forward direction. The goal is to collect fish bones while avoiding obstacles and enemies. The game progressively increases in difficulty.

---

## 5. STORY AND CHARACTERS

### 5.1 Main Character: Trash Cat

Trash Cat is a survival-driven character that searches for food in a harsh environment. It is agile and fast, making it suitable for an endless running gameplay style.

### 5.2 Enemy: Dog

The dog acts as an antagonist. It creates obstacles and pressure on the player, increasing the challenge.

### 5.3 Storyline

The game is set in an environment where resources are limited. Trash Cat must continuously run and collect fish bones to survive while avoiding a hostile dog. The story is simple and supports fast-paced gameplay.

---

## 6. GAMEPLAY DESIGN

### 6.1 Core Mechanics

- Continuous forward movement
- Player-controlled lateral movement
- Jump and dodge system
- Obstacle avoidance
- Collectible items

## 6.2 Controls

- Keyboard (PC): Arrow keys
- Mobile: Touch / swipe input

## 6.3 Game Loop

1. Game starts
  2. Player runs automatically
  3. Obstacles spawn dynamically
  4. Player collects items
  5. Difficulty increases
  6. Game ends on collision
  7. Score is saved
- 

# 7. GAME SYSTEMS

## 7.1 Movement System

The character continuously moves forward using a constant velocity system. Player input affects lateral movement and jumping.

## 7.2 Spawn System

The environment is divided into segments. These segments are spawned ahead of the player and destroyed behind the player to optimize performance.

## 7.3 Collision System

Collision detection is handled using Unreal Engine collision components. When the player collides with an obstacle, the game ends.

## 7.4 Scoring System

The score increases based on:

- Distance traveled
- Number of collectibles gathered

## 7.5 High Score System

High scores are stored using Unreal Engine's Save Game system, which stores data internally without requiring external storage permissions.

---

## 8. ASSET PIPELINE

### 8.1 Asset Source

Assets were obtained from Unity Asset Packs.

### 8.2 Conversion Process

1. Assets imported into Unity
2. Exported using FBX Exporter
3. Imported into Unreal Engine
4. Materials reassigned manually

### 8.3 Tools Used

- Unity (asset extraction)
  - Unreal Engine (development)
  - Blender (optional editing)
- 

## 9. DEVELOPMENT TOOLS

- Unreal Engine 5
  - Visual Studio (C++ support if needed)
  - Android Studio (SDK/NDK setup)
  - Blender
  - Photoshop / GIMP
- 

## 10. TECHNIQUES IMPLEMENTED

### 10.1 Procedural Generation

Track segments are generated dynamically to create an infinite environment.

### 10.2 Object Management

Unused objects are removed to maintain performance.

### 10.3 Mobile Optimization

- Reduced polygon count
- Optimized textures
- Efficient memory usage

### 10.4 Data Persistence

Unreal Save Game system is used to store high scores.

---

## 11. USER INTERFACE DESIGN

### UI Components

- Main Menu
- Gameplay HUD
- Game Over Screen

### Features

- Portrait orientation support
  - Responsive layout for mobile devices
- 

## 12. PLATFORM CONFIGURATION

### Android

- Minimum SDK: Android 8.0
- Target SDK: Android 14
- Internal storage usage

### Windows

- Compatible with Windows operating systems
- 

## 13. TESTING AND DEBUGGING

### Testing

- Manual playtesting
- Device testing (Android)

### Debugging Tools

- Unreal Engine logs
  - Packaging logs
- 

## 14. CHALLENGES AND SOLUTIONS

### Challenge 1: Android SDK Setup

Solution: Installed required SDK, NDK, and build tools.

### Challenge 2: Storage Permission Issue

Solution: Used internal storage instead of external storage.

### **Challenge 3: Asset Conversion**

Solution: Used FBX export pipeline.

### **Challenge 4: Performance Optimization**

Solution: Implemented object cleanup and efficient spawning.

---

## **15. FUTURE ENHANCEMENTS**

- Add sound effects and music
  - Improve enemy AI behavior
  - Add power-ups
  - Implement leaderboard system
- 

## **16. CONCLUSION**

The Trash Cat Endless Runner project successfully demonstrates the implementation of core game development concepts using Unreal Engine. The project highlights skills in gameplay design, system implementation, asset integration, and cross-platform deployment. It serves as a strong foundation for further development and advanced game projects.

---

## **17. REFERENCES**

- Unreal Engine Documentation
  - Android Developer Documentation
  - Unity Asset Store Resources
- 

**END OF DOCUMENT**